The FREUID Challenge 2026

Technical Report — test_freuid

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Abstract

We detect fraudulent identity documents (physical manipulations, GenAI-generated digital edits, and print-and-capture attacks) with a single fine-tuned ConvNeXt-V2-large classifier. The key finding driving our result was that FREUID Score is dominated by input resolution and by preserving native document aspect ratio: replacing square-padded low-resolution inputs with an aspect-preserving rectangular resize and progressively increasing training/inference resolution improved our public FREUID Score by roughly two orders of magnitude (square 384px: 0.198 → rect 832px, no TTA: 0.00465), without any architectural ensembling. We additionally applied capture-style augmentation (perspective jitter, re-compression, sensor noise, tone/contrast jitter) to close the domain gap between the training set (69,332 digital vs. only 20 physically captured images) and a test distribution that emphasizes print-and-capture documents. Diverse-architecture ensembling (a normalizer-free NFNet) and multi-seed averaging of the same recipe were both tried and both *hurt* the public leaderboard score once validated, despite promising offline agreement statistics; we report these as negative results. Our selected submission scores **0.00465** (FREUID Score, lower is better) on the public leaderboard. Code, the frozen model checkpoint, and a network-isolated Docker inference sandbox are provided for full reproducibility.

Competition: The FREUID Challenge 2026 (IJCAI-ECAI 2026) [1]
Kaggle team: test_freuid
Code: https://github.com/[TODO:your-org]/[TODO:your-repo] (commit [TODO: SHA])

1 Introduction

Identity document fraud detection must generalize across three qualitatively different attack surfaces: (i) physical manipulation of a genuine document (reprints, overlays, tampered security features), (ii) fully digital, GenAI-assisted edits (text/face/layout changes), and (iii) print-and-capture attacks, where a manipulated or synthetic document is physically printed and re-photographed, destroying many of the pixel-level artifacts a purely digital detector would rely on. The official metric, the FREUID Score, combines AuDET with APCER at a fixed 1% BPCER operating point (lower is better).

Our approach is intentionally simple: a single global-context vision backbone, fine-tuned end-to-end at high, aspect-preserving input resolution, with augmentation designed to simulate the print-and-capture domain gap present in the released training data. We favored this over a multi-branch/fusion architecture (global + patch-anomaly + OCR/layout + physical-artifact branches) after observing that resolution and aspect-ratio handling alone closed most of the gap to a strong baseline, and that adding a second, architecturally diverse but weaker branch strictly hurt validated leaderboard performance (Section 5).

2 Method

2.1 Overview

The pipeline is: (1) decode the document image with OpenCV (libjpeg-turbo), falling back to Pillow only if OpenCV cannot read the file; (2) resize with an aspect-preserving rectangle (no padding); (3) run the image through a fine-tuned ConvNeXt-V2-large backbone with a linear binary head; (4) apply a sigmoid to obtain a fraud score in $[0, 1]$. No post-hoc score calibration, temperature scaling, or thresholding is applied — the raw sigmoid output is submitted directly, matching the leaderboard's numeric semantics (higher = more confident the document is fraudulent).

2.2 Model architecture

Backbone: `convnextv2_large` [3] (timm [2]), 1536-dim pooled feature output, average pooling, **no frozen layers** (full end-to-end fine-tune). Head: `Dropout(p=0.1)` → `Linear(1536, 1)` producing a single logit. Total forward pass is a single backbone invocation per image; no test-time augmentation, no ensembling in the selected submission. Inference batch size is 1 (see Section 4 for why this does not affect numerics here).

2.3 Training objective and procedure

We fine-tune from ConvNeXt-V2's public ImageNet-pretrained weights (loaded once at training time only — the released checkpoint we submit contains the fully fine-tuned weights and requires no further downloads). Loss: focal loss ($\gamma = 2.0$, $\alpha = 0.75$) to handle the label imbalance (40,005 genuine vs. 29,347 fraudulent in the training set). Optimizer: AdamW with two learning-rate groups (backbone 1×10^{-5}, head 1×10^{-4}), weight decay 1×10^{-4}, gradient clipping, and a `OneCycleLR` schedule with linear warmup. We train for a single epoch over the *entire* labeled training set (no held-out fold, i.e. `-fit_all`) once the recipe was locked in via earlier fold-based experimentation, in order to maximize the data available to the final model. Effective batch size 8 (`batch_size=1`, `accumulate_grad=8`) — required at our largest input resolution to fit a 16 GB GPU.

Augmentation. Standard augmentations: horizontal flip, affine (translate/scale/rotate), motion/Gaussian blur, Gaussian noise, brightness/contrast/hue/saturation jitter, JPEG re-compression, coarse dropout. *Capture-style* augmentation (enabled via `-strong_capture_aug`) additionally applies: perspective jitter (simulating an off-axis phone/scanner capture), a random choice of downscale-then-upscale or aggressive JPEG re-encoding (simulating re-photographing a document), sensor-realistic noise (Gaussian or ISO noise), and gamma/CLAHE/brightness-contrast jitter (simulating varying capture lighting). This was motivated directly by the training-set composition (Section 3).

2.4 Score calibration

None applied. The sigmoid output of the fine-tuned head is submitted as-is at both public and (planned) private evaluation; this matches how the public leaderboard score was obtained, so no train/test mismatch is introduced by a separate calibration step.

3 Data

3.1 FREUID training set

69,352 labeled images across five document types/classes (`EGYPT/DL`, `GUINEA/DL`, `BENIN/DL`, `MOZAMBIQUE/DL`, `MAURITIUS/ID`), label distribution 40,005 genuine / 29,347 fraudulent. Critically, the `is_digital` metadata field shows only **20** images are physically captured (recaptured) versus 69,332 purely digital images — while the held-out/public test distribution emphasizes print-and-capture documents much more heavily. This digital → captured domain gap, not document type, is the dominant generalization challenge in this competition (see the discussion log, `discussion.txt`, in the repository for the full diagnostic trail that led to this conclusion). We used capture-style augmentation (Section 2) as our primary mitigation; a follow-up experiment oversampling the 20 real captured rows within each training epoch was run late in development — [**TODO: report its validated public-LB outcome here once confirmed; see repository memory/discussion log for the in-progress result at report-drafting time**].

3.2 External data and pre-trained models

No external data beyond FREUID. The only external resource used is the publicly available, permissively licensed ImageNet-pretrained `convnextv2_large` checkpoint distributed via `timm` (downloaded once during training only). The released, fully fine-tuned inference checkpoint requires *no* network access and is included in this repository (see `models/README.md`).

3.3 Validation strategy

Early architecture/resolution/augmentation decisions were validated with a group-stratified out-of-distribution-by-document-type fold (`StratifiedGroupKFold` on `type`). This fold strategy was useful for catching gross regressions but **underestimated the public-leaderboard digital/captured domain gap** (Section 3): capture-style augmentation that looked neutral-to-harmful on the OOD-by-type fold in fact improved the public leaderboard substantially, so later development decisions (input resolution, inference scale, capture augmentation strength, ensembling) were validated directly against the public leaderboard rather than the offline fold. Private test labels were, by construction, never available during model selection.

4 Inference

Test-time preprocessing matches training preprocessing except augmentation is disabled: aspect-preserving rectangular resize to height 736 (width $= \mathrm{round}(736 \times 1.585/32) \times 32 = 1152$), OpenCV decode with a Pillow fallback, ImageNet normalization. **No test-time augmentation** is applied in the selected submission (an earlier flip-averaged TTA variant was strictly worse once the decode path was corrected — see Section 5). Inference resolution (736) is deliberately *below* training resolution (832, $\approx 0.88\times$); this scale was found empirically to sit at the minimum of a noisy but clearly unimodal inference-scale response curve (704 / 736 / 768 / 800 / 832 all evaluated). Batch size is 1; since ConvNeXt-V2 uses LayerNorm rather than BatchNorm, batch size does not affect numerics at eval time, so this choice is purely about matching the exact validated configuration and about being able to attribute a decode failure to a single image in the Docker sandbox (see `docker/README.md`). The exact command used to produce the selected submission via the CSV workflow, and the equivalent Docker entrypoint used for the private/final evaluation, are both documented in the repository's top-level `README.md` and `docker/README.md`.

Hardware: single 16 GB GPU (16,376 MiB), WSL2/Linux, 54 GB RAM + 64 GB swap. Scoring the 7,821 publicly-scored rows takes roughly 15–18 minutes at scale 736 without TTA. [**TODO: measure and report CPU-only wall-clock, and private-set wall-clock once the private images and their count are available on/after July 13.**]

5 Results

Private leaderboard: [**TODO: fill in after the private test images are released (July 13, 2026) and the final submission is scored, per the code freeze in Section 6**].

Table 1: Selected leaderboard results (lower FREUID Score is better).

Configuration	FREUID ↓	Notes
Square-pad, 384px	0.198	Early baseline
Rect (aspect-preserving), 448px	0.16050	
Rect, 512px	0.15403	
Rect, 640px	0.05149	Step-change vs. 512px
Rect, 768px	0.02342	
Rect, 832px, infer @768, TTA	0.00515	
Rect, 832px, infer @736, TTA	0.00495	Previous champion
Rect, 832px, infer @736, **no TTA**	**0.00465**	**Selected submission**
NFNet (diverse arch.) standalone @736	0.11438	Weaker; any ensemble weight on it hurt
ConvNeXt/NFNet ensemble (best weight)	0.00538	Still worse than ConvNeXt alone
Two-seed ConvNeXt average (same recipe)	0.00685	Hurt despite high pairwise agreement (r=0.936)

5.1 Negative results

We highlight two results that looked promising by offline proxy metrics but did not improve the validated (public leaderboard) score, since we believe they are informative for other teams:

- **Diverse-architecture ensembling.** A normalizer-free NFNet (`dm_nfnet_f1`), trained with an otherwise identical recipe, reached a competitive training loss but scored 0.11438 standalone on the public leaderboard — over 20× worse than the ConvNeXt-V2 champion. Every non-zero ensemble weight on the NFNet branch monotonically *increased* (worsened) the FREUID Score. We conclude the public test distribution (skewed toward print-and-capture images) is far outside this architecture's effective generalization, despite good in-distribution training loss.
- **Multi-seed averaging.** Two independently-seeded runs of the identical champion recipe agreed strongly offline (Pearson $r = 0.936$, 95.0% sign agreement at the 0.5 threshold on the 7,821 publicly-scored rows) — a much stronger agreement signature than the NFNet pair, and the textbook precondition for variance-reducing ensembling. Nonetheless, both a 50/50 average (0.00685) and small-weight blends (5%: 0.00507, 10%: 0.00510) scored *worse* than the single champion checkpoint alone (0.00465), and the second seed's standalone score (0.03906) turned out to be far weaker than the first despite comparable training loss. We did not find a reliable offline signal that distinguished this failure case from a genuinely beneficial ensemble ahead of time.

6 Reproducibility

- **Repository:** `https://github.com/[TODO:your-org]/[TODO:your-repo]`, commit `[TODO: 40-character SHA, frozen at or after the July 13 private-test release per the competition's code-freeze policy]`.
- **Docker:** 'docker build -f docker/Dockerfile -t freuid-repro:local .' (build context = repository root); weights are baked into the image from `models/convnext_full_drop01_rect832_capaug_e1.pth` (SHA-256 in `models/README.md`) — no runtime downloads. Run with `docker run -network none -v /data:/data:ro -v /out:/submissions freuid-repro:local`. See `docker/README.md` for the full contract and environment-variable configuration.
- **Dependencies:** Python 3.11; see `docker/requirements.txt` for pinned, broadly-installable version ranges, and the top-level `README.md` for the exact development-environment versions (`torch 2.11.0+cu126`, `timm 1.0.27`, `opencv-python 4.13.0`, `albumentations 2.0.8`).

- **Runtime:** single 16 GB GPU recommended; CPU fallback is supported but not benchmarked in this draft — [**TODO: fill in CPU wall-clock before the final submission**].
- **Randomness:** training used a fixed seed (42) via `src/utils/seed.py::seed_everything`; inference is deterministic given the frozen checkpoint (no dropout/augmentation at eval time, no TTA in the selected submission).
- **Network:** inference requires *no* network access (`docker run -network none`); network is only used at `docker build` time to install Python dependencies.

7 Team and contributions

Solo entry. Michael Semenoff (team test_freuid) performed all modeling, experimentation, data analysis, engineering (training pipeline, Docker reproducibility sandbox), and writing.

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References

[1] Ivan Relić, Vincenzo D'Elia, Stefano Bortoli, Radu Tudoran, Hristina Nedyalkova, Mihaela Bošnjak, Stanislav Pavlić, Tin Mavračić, Darin Dašić, Marin Kačan, Jerko Šegvić, Paolo Čerić, Filip Šoprun, Matej Miočić, Marin Oršić, Lorenzo Vaiani, and Paolo Garza. The FREUID challenge 2026: 1st competition on identity documents fraud detection. https://freuid2026.microblink.com/, 2026. IJCAI-ECAI 2026 competition.

[2] Ross Wightman. PyTorch image models. https://github.com/huggingface/pytorch-image-models, 2019.

[3] Sanghyun Woo, Shoubhik Debnath, Ronghang Hu, Xinlei Chen, Zhuang Liu, In So Kweon, and Saining Xie. ConvNeXt V2: Co-designing and scaling convnets with masked autoencoders. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

A Hyperparameters

B Additional ablations

See Table 1 and the negative-results discussion in Section 5. The full, chronological experimental log (including every intermediate resolution/scale/augmentation configuration tried) is preserved in `discussion.txt` in the repository for transparency.

Hyperparameter	Value
Backbone	`convnextv2_large` (timm), full fine-tune
Training image size	832 (rect, aspect 1.585 $\Rightarrow$ 832$\times$1312)
Inference image size	736 (rect, aspect 1.585 $\Rightarrow$ 736$\times$1152)
Batch size (train)	1 (accumulate_grad = 8, effective 8)
Batch size (inference)	1
Learning rate (head)	1×10^{-4}
Learning rate (backbone)	1×10^{-5}
Weight decay	1×10^{-4}
Head dropout	0.1
LR schedule	OneCycleLR, linear warmup
Loss	Focal loss ($\gamma = 2.0$, $\alpha = 0.75$)
Epochs	1 (full training set, no held-out fold)
Seed	42
Test-time augmentation	None (selected submission)